MTH 446: Lecture # 5

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Lecture Span

1. Enter here

Review

To evaluate $\sqrt{1-i\sqrt{3}}$, we first find the $re^{i\theta}$ representation of this complex number, then take the root of this representation, which is the following:

$$r^{1/2}\exp(i\theta/n + 2\pi k/n)$$

.

For instance,

$$2\left(\frac{1}{2} - i\frac{\sqrt{3}}{2}\right) = \cos\left(-\pi/3\right) + i\sin(-\pi/3) \tag{1}$$

$$= 2\exp\left(-i\frac{\pi}{3}\right) \tag{2}$$

Therefore,

$$c_0 = \sqrt{2}\exp\left(-i\frac{\pi}{6}\right) = \sqrt{2}\left(\cos(-\pi/6) + i\sin(-\pi/6)\right) \tag{3}$$

$$= \sqrt{2}\left(\frac{\sqrt{3}}{2} - i\frac{1}{2}\right) \tag{4}$$

$$= \frac{\sqrt{6}}{2} - i\frac{\sqrt{2}}{2} \tag{5}$$

Moreover, $c_1 = -c_0$, therefore,

$$c_1 = -\frac{\sqrt{6}}{2} + i\frac{\sqrt{2}}{2} \tag{6}$$

Regions in the complex plane

Definition 0.1. *$\underline{\epsilon\text{-neighborhood}}$. Let $\epsilon > 0$ and $z_0 \in \mathbb{C}$, then the $\underline{\epsilon\text{-neighborhood of } z_0}$ is defined as:*

$$\mathcal{D}_\epsilon = \{z \in \mathbb{C} : |z - z_0| < \epsilon\} \tag{7}$$

This defines a 2D circle on the complex plane where all values in $\mathcal{D}_\epsilon$ are contained within a neighborhood of ϵ around the point z_0.

Definition 0.2. *$\underline{\text{Deleted } \epsilon\text{-neighborhood}}$. Let $\epsilon > 0$ and $z_0 \in \mathbb{C}$, then the $\underline{\text{deleted } \epsilon \text{ neighborhood}}$ is defined as*

$$\dot{\mathcal{D}}_\epsilon = \{z \in \mathbb{C} : 0 < |z - z_0| < \epsilon\} \tag{8}$$

The region of $\dot{\mathcal{D}}_\epsilon$ doesn't include the point z_0,, and is a subset of $\mathcal{D}_\epsilon$.

Definition 0.3. *Let $U \subset \mathbb{C}$ and $z_0 \in U$. Then the point z_0 is an* <u>*interior point*</u> *of U if there exists $\epsilon_0 > 0$ such that*

$$\mathcal{D}_\epsilon(z_0) \subset U \tag{9}$$

This means that there exists a small circle around z_0 if z_0 is in U. This excludes any points on ∂U.

Definition 0.4. *Let $U \in \mathbb{C}$ and $z_0 \notin U$. Then the point z_0 is called the* <u>*exterior point*</u> *of U if $e_0 > 0$ such that*

$$\mathcal{D}_\epsilon \subset \mathbb{C}/U \tag{10}$$

The inverse of the previous definition, minus the boundary points of U.

Definition 0.5. *Let $U \subset \mathbb{C}$ and $z_0 \in \mathbb{C}$. Then the point z_0 is called a* <u>*boundary point*</u> *of U if z_0 is neither an interior point or an exterior point.*

The set of all boundary points of U is denoted by ∂U

So, $z_0 \in \mathbb{C}$ is a boundary point of U (i.e., $z_0 \in \partial U$). Therefore, $z_0 \in \partial U$ if and only if z_0 is not an interior point and z_0 is not an exterior point.

z_0 is an interior point of U if and only if $\exists \epsilon_0 > 0$ such that $\mathcal{D}_{\epsilon_0} \subset U$.

z_0 is not an interior point of U if and only if

$$\neg(\exists \epsilon_0 : \mathcal{D}_{\epsilon_0}(z_0) \subset U) \iff \forall \epsilon > 0,\ \mathcal{D}_\epsilon \cap \mathbb{C}/U \neq \emptyset \tag{11}$$

Then, z_0 is an exterior point of U if and only if

$$\exists \epsilon_0 > 0 : \mathcal{D}_{\epsilon_0}(z_0) \subset \mathbb{C}/U \tag{12}$$

Then, z_0 is not an exterior point of U if and only if

$$\forall \epsilon > 0, \mathcal{D}_\epsilon \cap U \neq \emptyset \tag{13}$$

Therefore, $z_0 \in \partial U$ if and only if (z_0 is not an interior point) and (z_0 is not an exterior point) if and only if **This statement is really important!**

$$\forall \epsilon > 0, (\mathcal{D}_\epsilon(z_0) \cap \mathbb{C}/U \neq \emptyset) \wedge (\mathcal{D}_\epsilon(z_0) \cap u \neq \emptyset) \tag{14}$$

In words, $z_0 \in \mathbb{C}$ is a boundary point of U if and only if $\forall \epsilon > 0$, $\mathcal{D}_\epsilon(z_0)$ contains points in U and outside of U.

Definition 0.6. *A set $U \subset \mathbb{C}$ is called an open set if U contains no boundary points if and only if $\partial U \cap U \emptyset$.*

1. *Notation: $U \subset \circ\mathbb{C} \iff U$ is open in $\mathbb{C}$*

We claim that the emptyset $\emptyset$ is open in $\mathbb{C}$. This also means that $\emptyset \subset \circ\mathbb{C} \iff \partial\emptyset \cap \emptyset = \emptyset$

Something to note: If $u \neq \emptyset$, then $U \subset \circ\mathbb{C} \iff$ every point of U is an interior point. Therefore, a non-empty set $U \subset \mathbb{C}$ is open if and only if for every $z_0 \in U$, then there is an $\epsilon > 0$ such that $\mathcal{D}_\epsilon \subset U$.

In particular, $\mathbb{C}$ is clearly open.

Proof. We wish to prove that

$$(u \neq \emptyset) \wedge (u \subset \circ\mathbb{C}) \iff (u \neq \emptyset) \wedge (\text{every point of U is interior}) \tag{15}$$

In other words, given $U \neq \emptyset$ and $U \subseteq \circ\mathbb{C}$, then we wish to prove that every point of U is an interior point. Firstly, let $z_0 \in U \implies z_0 \notin \partial U$. Then

$$z_0 \notin \partial U \iff \neg(z_0 \text{ is not an interior point} \wedge z_0 \text{ is not an exterior point}) \tag{16}$$

Therefore, z_0 is an interior point or an exterior point. But also z_0 is in U. Therefore, z_0 cannot be an exterior point. Therefore, z_0 is an interior point. □

Proof. We next prove ($\impliedby$). Here, we are given that every point in U is an interior point. We wish to prove that U is open in $\mathbb{C}$.

Let $z_0 \in U$, then if z_0 is an interior point, then there exists an $\epsilon > 0$ such that $\mathcal{D}_\epsilon \subseteq U$. But this is the definition of an open set. □

Definition 0.7. *A set $F \in \mathbb{C}$ is* <u>*closed*</u> *if $\partial F \subseteq F$.*

Examples

We prove that $\partial\emptyset = \emptyset$. Contrary to the claim that $\partial\emptyset \neq \emptyset$. Then $\exists z_0 \in \partial\emptyset \iff \forall(\mathcal{D}_\epsilon \cap \emptyset \neq \emptyset) \wedge (\mathcal{D}_\epsilon \cap \mathbb{C}/\emptyset \neq \emptyset)$. However, the intersecton of any set with the empty set is empty. Therefore, we arrive at a contradicton.

Therefore, we conclude that $\emptyset \subseteq \circ\mathbb{C}$ because $\partial\emptyset = \emptyset$ and $\partial\emptyset \cap \emptyset = \emptyset$. But also, $\partial\emptyset \subseteq \emptyset$, therefore, the empty set is also closed.

Next, we prove that $\partial\mathbb{C} = \emptyset$. For the sake of contradiction, suppose $\partial\mathbb{C} \neq \emptyset$, then $\exists z_0 \in \partial\mathbb{C} \iff \forall(\mathcal{D}_\epsilon \cap \mathbb{C} \neq \emptyset) \wedge (\mathcal{D}_\epsilon \cap \mathbb{C}/\mathbb{C} \neq \emptyset)$. But $\mathbb{C}/\mathbb{C} = \emptyset$, therefore, we arrive at a contradiction.

We can also prove that $\mathbb{C}$ is a open set. But since $\emptyset \subseteq \mathbb{C}$, $\mathbb{C}$ is also a closed state.

Proposition 0.8. *A set F is closed if and only if $\mathbb{C}/F$ is open.*

We will prove it next time.